

user as well as asking simple questions occasionally to understand his likes and dislikes, judge his mood, etc. In order to avert concerns about the user's privacy, they use smart camera networks that process the images locally to abstract the required information and delete them promptly.

Their experiments also involve interactive machine learning. That is, the devices in the room try to observe the user, understand his needs and adapt their behaviour accordingly. But how well have these devices adapted to the user's needs? This is something that the user alone can judge. The devices interact with the user to let him judge how well the adaptation or reactions are, and recalibrate if needed.

Media convergence. There is a great potential for media convergence in an ambient intelligence scenario. The many tech tools in a home or office can be integrated into a user-centric system, which one can employ to access data, multimedia, music, news, e-mail, Internet content, as well as control the living and work environment parameters such as lighting, air-conditioning, ambiances, appliances and media centres at home, collaborative spaces, shared documents, gesture-controlled presentations and video conferencing at work.

The convergence of a plethora of devices and media will also encourage better understanding of the user. Devices such as mobile phones, notebook computers, automobile information centres, home entertainment centres

Opportunities in ambient electronics

- RFID
- Wi-Fi
- Bluetooth
- Near-field communications
- Home-area networking
- Embedded electronics
- Board (PCB) development
- Embedded operating systems and software
- Voice recognition
- Internet infrastructure (including cloud technologies)
- Flat displays – LCDs
- LEDs
- System-in-package

—Suggested by S. Uma Mahesh

and personal game consoles can all be considered as interfaces to acquire and embed user preferences, which, in turn, can act as contextual data in adapting the services rendered by different ambient intelligence applications.

Devices so tiny and flexible that you don't even notice them. One of the prerequisites for ambient electronics is that the devices should be so small and flexible that they blend with the environment. You will notice them only by their action, and not by their presence. This requires not just small and capable chips but also inconspicuous electronics packaging. Nano-electronics and organic electronics hold a lot of promise in this respect.

Organic electronics is essentially made of conductive polymers, plastics or small molecules. Like the molecules of living things, it is based on carbon (not inorganic conductors like copper or silicon) and hence organic. Organic or polymer electronics includes thin-film or foil-like electronics, transparent electronic packages and paper-based electronics. The technology will play an important role in memories, smart clothing, radio-frequency identification (RFID) tags, chemical sensors, solar cells and displays, all of which are essential aspects of ambient intelligence. In short, organic electronics would make it possible to convert a book, wall or paper

napkin into a device that could help you study, work or eat.

Organisations like the IMEC (Belgium), the Max Planck Institute and IBM are working on developing organic electronic products.

Smarter vision. Computer vision systems are very essential for observing and understanding the user—a key requirement for the adaptive and context-aware behaviour of ambient electronics. Today's vision systems and networks are quite bulky and inflexible, not quite qualifying as ambient electronics.

Organisations like IMEC are working to develop smart vision systems that have better flexibility, speed and accuracy by using components that integrate the micro-electromechanical systems (MEMS) and driving logic on one system.

Carnegie-Mellon University's (CMU) Instinctive Computing Lab also does a bit of good work in this space. A research team there realised that the main bottleneck of today's video surveillance systems is that we have too much information but not enough network bandwidth, attention and intelligence. Over 99 per cent of data fed into the control room is wasted. So they undertook to minimise the wireless video network throughput, while maximising the video quality with multiple resolutions on demand. "This would enable detection of events or features in the video," they say. Their Cognitive Agent Network applies real-time eye-gaze-tracking technology to detect where the user looks at and switch the resolution accordingly.

They report, "From our empirical experiments, we found that multiple resolution screen switching can reduce the network traffic by about 39 per cent. Manual switching of the four screens can reduce the data flow by about 57 per cent. With eye gazing interface, we obtain about 75 per cent reduction of the network traffic."

Another key concern in vision systems is user privacy. Recent research has made it possible to use 3D millimetre-wavelength imaging systems

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